

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15A.3

OPTICAL ACTIVITY AND REACTION MECHANISMS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15A.3: Optical Activity and Reaction Mechanisms
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	6 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15A.3

Q1. 1. (a) Nucleophilic substitution of (R)-2-bromobutane with aqueous KOH produces a racemic mixture of butan-2-ol. Deduce whether this reaction proceeds via SN1 or SN2 mechanism and explain your reasoning.

[4 Marks]

WCH14/01/Oct22/Q6(a)

Q2. (b) Contrast the optical activity of the product formed when (R)-2-bromobutane undergoes SN2 nucleophilic substitution.

[2 Marks]

WCH14/01/Oct22/Q6(b)

SECTION B: WORKED MARK SCHEME — 15A.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Nucleophilic substitution of (R)-2-bromobutane with aqueous KOH produces a racemic mixture of butan-2-ol. Deduce whether this reaction proceeds via SN1 or SN2 mechanism and explain your reasoning. 1. SN1 mechanism. 2. SN1 involves formation of a planar carbocation intermediate. 3. The hydroxide nucleophile can attack the planar carbocation with equal probability from above or below the plane. 4. This forms equal amounts of both enantiomers (50:50 racemic mixture).	[4]
Q2	Q2: (b) Contrast the optical activity of the product formed when (R)-2-bromobutane undergoes SN2 nucleophilic substitution. 1. SN2 involves backside attack in a single concerted step. 2. Causes inversion of configuration (Walden inversion), yielding a single optically active enantiomer.	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15A.3**FAQ #1: How does optical activity prove the mechanism of nucleophilic substitution?****Category: Mechanisms & Evidence****COMMON EXAMINER TRAP / MISTAKE:**

Confusing SN1 (planar carbocation → racemic / inactive) with SN2 (inversion → optically active).

MODEL ANSWER / EXAMINER EXPECTATION:

SN1 produces a trigonal planar carbocation intermediate allowing attack from top/bottom equally → racemic mixture (optically inactive). SN2 proceeds via single-step backside attack → inversion of configuration → single optically active product.